Lesson plan · for the teacher

Data ethics: fair, safe and honest

IoT · Level 4 · Lesson 4

Time
40 min

Materials
The board and chalk, or large paper and markers; A four-column table: need it?, could reveal, who sees, delete when; Paper for the families' notice; Optional: the class's one-page plan from lesson 1

Learning goals

- Decide what data a project really needs, and what it could reveal about people.
- Plan consent, access and how long data is kept, with extra care for children.
- Share results honestly with the school community, uncertainty included.

Activity: Data audit

1. Warm-up (5 min): Tell the story of the CO_2 sensors that showed one room empty most afternoons, and the head teacher who asked whose room it was. Ask: what was the data about?
2. Teach (8 min): What sensor data can reveal; collect only what the question needs; tell people and ask for consent; who sees what and for how long; honest reports with their limits.
3. Audit (12 min): Groups take the class's project (or a made-up one) and fill a four-column table for each piece of data: need it?, could reveal, who sees, delete when.
4. Cut (5 min): Groups cross out every piece the question doesn't need, and say why in one sentence.
5. Notice (10 min): The class writes a five-line notice for families: what is collected, why, who sees it, how long it is kept, and whom to ask.

Use a planned or made-up project, never real pupils' data. Keep the discussion about practices, not colleagues: no names.

Discussion

- What surprised you about what sensor data can reveal?
- Who should decide when project data is used for something new?
- How would you report a result you're only half sure of?

Exit ticket

- Rewrite this claim honestly: “Room 6 has the worst air in the school.”

Try it at school tomorrow
Data audit: With your class, list every piece of data your project (or a planned one) collects. For each, pupils ask four questions: do we need it, what could it reveal, who sees it, and when is it deleted? Cross out what isn't needed, then write a five-line notice for families in plain words, in the language they speak at home.

Practice questions · for you

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1 **The head teacher asks the project team for the CO_2 data to see which teachers leave their rooms early. What should the team do?**

- ◯ Send only room averages, so no teacher is named
- ◯ Pause: a new use needs a new conversation and consent
- ◯ Hand it over: it's school data, and the head teacher is in charge

2 **A tracker on the school bus records its position every 10 seconds. What could this data reveal about people?**

- ☐ Which pupils sit next to each other
- ☐ Roughly where pupils live, from where the bus stops
- ☐ The driver's speed and breaks
- ☐ Who was absent, when a stop with one pupil is skipped
- ☐ Nothing: it's only about the bus

3 **Removing names from sensor data always makes it anonymous.**

◯ True ◯ False

4 **Put the steps of a fair data plan in order.**

- ☐ Tell families what, why, who sees it and for how long
- ☐ Delete the raw data at the agreed time, keeping only averages
- ☐ Collect, with access limited to the team
- ☐ Write the question the data must answer
- ☐ List only the data that question needs

5 **Match each concern to the practice that answers it.**

A report could blame a teacher	• • • •	A short notice in the home language
Families don't know what's collected	• • • •	Delete after a term, keep averages
Raw readings kept for years	• • • •	Ask again before using it
Results look surer than they are	• • • •	Describe the room's air, not the people in it
Someone proposes a new use	• • • •	State the sensors, dates and gaps

✎ **What data does your school already collect about pupils: registers, photos, WhatsApp groups? Which of it would you not want a stranger to see?**

Check your answers

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1 **Pause: a new use needs a new conversation and consent**

Staff and families agreed to a project about fresh air, not about being watched.

2 **Roughly where pupils live, from where the bus stops; Who was absent, when a stop with one pupil is skipped; The driver's speed and breaks**

Positions over time show stops, routes and timings, and so homes, absences and the driver's habits; they can't show what happens inside the bus.

3 **False**

CO_2 per room plus a timetable can point straight to one teacher, and a bus stop with one pupil points to one home.

4 **Top to bottom: 3, 5, 4, 1, 2**

The question decides what's needed; people are told before anything is collected; and deleting raw data at the agreed time keeps the promise you made at the start.

5 **A report could blame a teacher — Describe the room's air, not the people in it**
Families don't know what's collected — A short notice in the home language
Raw readings kept for years — Delete after a term, keep averages
Results look surer than they are — State the sensors, dates and gaps
Someone proposes a new use — Ask again before using it

Each practice answers one worry before it becomes a problem.